

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

May 2014

EE405 Power System Protection

Date: 23.05.2014, Friday

Time: 01:30 p.m. To 04:30 p.m.

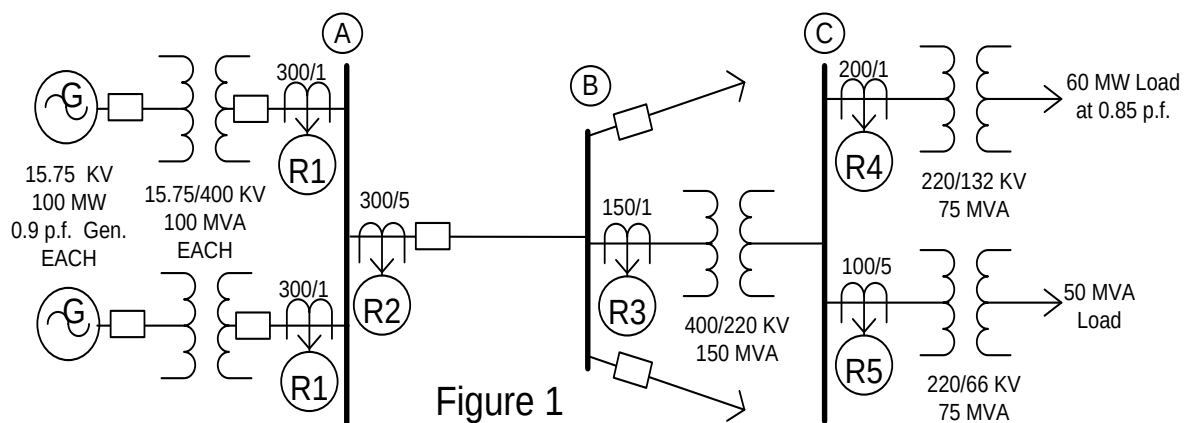
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** i. Instantaneous over-current relays suffer from the problem of over reach because [07]
_____ present in the fault current.
- ii. One disadvantage of using only time discrimination for transmission line protection is..... (Complete the sentence)
- iii. The over-current relay is to be monitored by under-voltage relay when..... (Complete the sentence)
- iv. Buchholz relay is provided for _____ Protection.
- v. Magnetising inrush current of transformers is rich in second harmonic contents. (state the sentence is true or false)
- vi. Mho relay is used for short transmission line protection. (state the sentence is true or false)
- vii. By burden of the relay, we generally mean VA rating of the relay. (state the sentence is true or false)
- Q - 2(a)** Figure 1 shows a single-line diagram of a portion of a power-system network. The [08]
relays shown are standard IDMT relays (setting range= 50 to 200% of CT secondary rating). The system permits continuous 10% overloading. The three phase short circuit current of bus A, B and C are 6000 Amp, 5600 Amp and 1900 Amp respectively. TMS of R_5 and R_4 are set as 0.1 each. The high set instantaneous setting of relay R_4 and R_5 are set at 800% and 2000% respectively. Determine the plug setting of all relays and time setting of R_1, R_2 and R_3 .



- (b) Distance Relays cannot be applied for the transmission lines employing auto reclosers at both the ends of the line. [03]
- (c) Give Reason: Over fluxing protection is provided for generator transformers only. [03]

OR

- Q - 2(a)** For the system shown in the Figure 2, all the relays are either directional or non-directional standard IDMT relays as per requirement. The fault level of bus A is 3500 MVA and that of bus B is 2500 MVA. Find the plug settings of the all Relays. Also find out the TMS of the all Relays if the TMS of R_2 and R_3 is set at 0.3 each. [08]

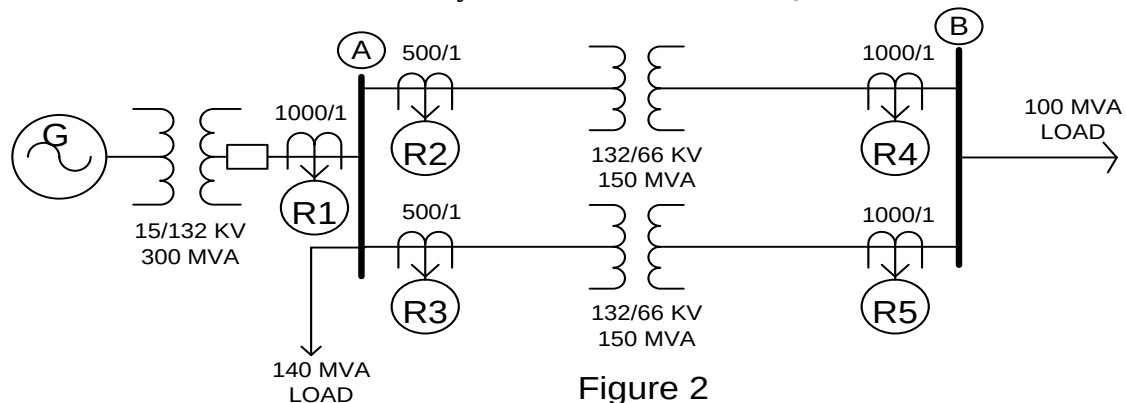


Figure 2

- (b)** For the power system shown in figure 3, complete the following table. [03]

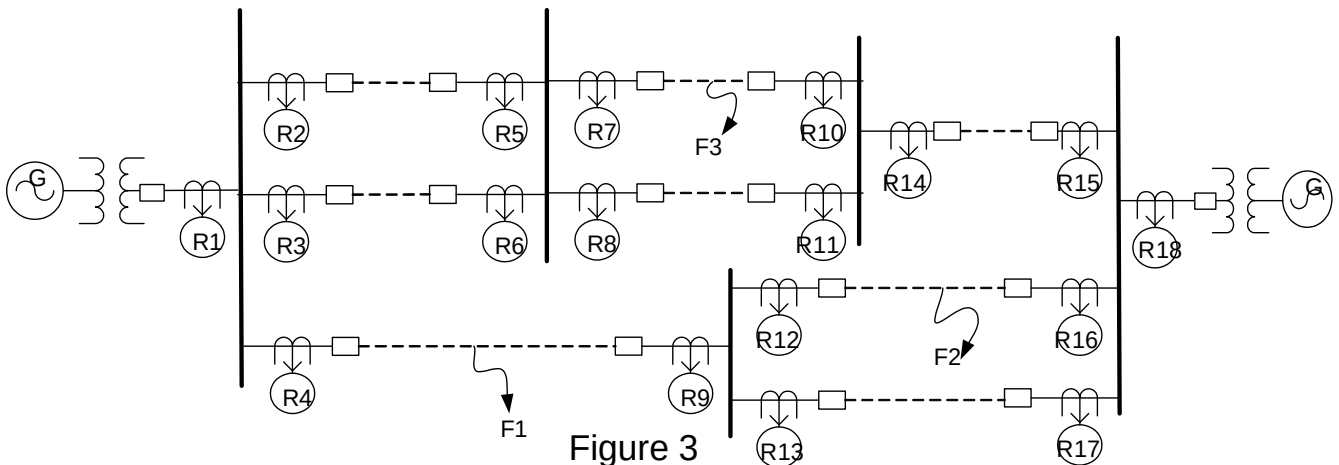


Figure 3

Fault at	Primary Relay fails	Backup relay/s operate
F1	R4	
F2	R16	
F3	R7	

- (c)** Give reason: In case of over-current protection of transmission line, there are four over-current relays required at one relaying point, where as for distance protection totally six distance relays required at one relaying point. [03]

- Q - 3(a)** From the following data, determine in terms of secondary ohms, the first zone setting of a reactance relay. Line section AB is to be protected, the positive sequence impedance of which is $3+j7.5$ ohms (primary). [08]

CT ratio=400/1

PT ratio=220KV/110V

The first zone covers 80% of the first line section. The second zone covers 150% of the line section AB. Both zones are protected by reactance characteristic. Both these characteristic are bound by using impedance relay for the third zone protection which covers up to 220% of AB. Find all three zone setting.

- (b)** Draw the block diagram of carrier current protection scheme and explain each block as well as component. [06]

OR

- Q - 3(a)** Explain in detail with suitable vector diagram: Inherent phase shift of currents in the transformer, which becomes obstacles in transformer differential protection. **[08]**
- (b)** Give reason: Instantaneous over current relays suffer from the problem of over reach. **[03]**
- (c)** Explain: For protection against phase faults, 90° connection is used, in case of directional relays. **[03]**

SECTION – II

- Q - 4(a)** Explain the reasons for using additional auxiliary contacts in Trip circuit of circuit breaker. **[02]**
- (b)** Explain the operation of Negative Phase sequence Relay with necessary diagrams. **[05]**
- Q - 5(a)** Explain the reason why we need to identify the fault even on the Neutral of Generator? Why portion of the winding remains un protected with a normal Earth fault protection? Explain the 100% Earth Fault Protection scheme to protect the Stator of Generator with necessary diagrams. **[07]**
- (b)** An 11 KV, three phase, 30 MVA, star connected alternator is protected by an earth fault relay having 10% setting. if the neutral resistance limits the maximum earth fault current to 40% of full load value, determine the value of the resistor and percentage of the winding protected. Find also the value of the earth resistor needed to allow only 9.5% of the winding to be left unprotected. CT ratio is 2000/1 Amp. **[07]**

OR

- (b)** With suitable diagrams explain the generator stator inter turn fault protection scheme. **[07]**
- Q - 6(a)** List out and explain in short: Abnormalities arises in induction motor. **[06]**
- (b)** List out and explain in short: Type tests that relay has to undergo. **[08]**

OR

- Q - 6(a)** Explain Frame Leakage Protection for busbar. **[07]**
- (b)** Explain step by step method to investigate problem present in Three Phase Over Current + E/F Protection Scheme with help of Primary injection method. **[07]**
